

MINI PROJECT SYNOPSIS
ON
DRINK WATER REMAINDER

Bachelor of Computer Applications
to
Uttaranchal School of Computing Sciences



SESSION (2025-26)

Under the supervision of
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DECLARATION

I hereby declare that the project report titled “**DRINK WATER REMAINDER**” was submitted Aarav Chauhan to Uttarakhand School of Computing Sciences. The project was done under the guidance of **Mr. Aagman Kaparwan**. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this university or any other university or institute.

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CERTIFICATE OF ORIGINALITY

This is to certify that the project entitles “**DRINK WATER REMAINDER**” by **Aarav Chauhan** has been submitted in the partial fulfilment of the requirements for the award of the degree of BCA from department of computer applications from Uttaranchal school of computing science **Uttaranchal University**, Dehradun.

The results embodied in this project have not been submitted to any other University or Institution for the record of any degree.

Under the guidance of:
Mr. Aagman Kaparwan
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1. INTRODUCTION

Water is one of the most essential components required for the survival of all living organisms. In the human body, water plays a vital role in maintaining body temperature, supporting digestion, transporting nutrients, removing waste products, and keeping organs functioning properly. The human body is composed of approximately 60% water, and maintaining an adequate level of hydration is essential for overall health and well-being.

In today's fast-paced lifestyle, many individuals become so busy with work, studies, or daily responsibilities that they forget to drink water regularly. This leads to dehydration, which may cause fatigue, headaches, reduced concentration, dizziness, and other health problems. In the long term, inadequate hydration can also affect physical performance and mental alertness.

To address this problem, the **Drink Water Reminder System** has been developed as a simple computer-based application using the **C++ programming language**. The purpose of this system is to help users maintain proper hydration by calculating the recommended amount of water intake based on their body weight and providing regular reminders to drink water.

The system works by taking input from the user such as **weight and age**, calculating the recommended daily water intake using a standard formula, and then generating reminders at user-defined intervals. The reminders are displayed on the screen along with a beep sound alert to ensure that the user notices the notification.

This project demonstrates the use of **basic programming concepts**, including variables, constants, loops, conditional statements, input/output operations, and system functions. It also illustrates how simple programming solutions can be applied to solve everyday health-related problems.

2. OBJECTIVES

The main objective of the Drink Water Reminder System is to encourage people to maintain proper hydration by reminding them to drink water regularly. The project aims to combine basic programming techniques with a real-life application that promotes healthy habits.

- **To promote healthy hydration habits:**

The main objective of the Drink Water Reminder System is to encourage users to drink water regularly throughout the day. Proper hydration is essential for maintaining good health, improving concentration, and supporting overall body functions.

- **To calculate recommended daily water intake:**

The system calculates the recommended amount of water a person should drink based on their body weight. This helps users understand how much water their body requires each day.

- **To provide regular reminder notifications:**

The program reminds users to drink water at specific intervals chosen by them. These reminders ensure that users do not forget to drink water during their busy schedules.

- **To implement sound alert notifications:**

The system generates a beep sound along with the reminder message so that the user can easily notice the alert even if they are not actively looking at the screen.

- **To apply basic C++ programming concepts:**

This project aims to demonstrate the use of fundamental C++ programming concepts such as loops, conditional statements, variables, constants, and input/output operations.

- **To create a simple and user-friendly system:**

The program is designed to be easy to use so that users can operate it without any technical difficulty.

3. HARDWARE AND SOFTWARE USED

➤ Hardware Requirements

- Computer / Laptop
- Minimum 2 GB RAM
- Processor: Intel i3 or above
- Keyboard and Monitor

➤ Software Requirements

- Programming Language: C++
- Compiler: **Turbo C++ / Dev C++ / CodeBlocks / Visual Studio**
- Operating System: **Windows**
- Libraries Used:
 - `iostream` for input and output
 - `Windows.h` for `Sleep()` and `Beep()` functions

4. SYSTEM ANALYSIS

System analysis is the process of studying the existing problem, understanding user requirements, and designing a solution that meets those requirements effectively. It involves identifying system inputs, processes, outputs, and constraints. The goal of system analysis is to develop a system that is efficient, reliable, and easy to use.

For the **Drink Water Reminder System**, system analysis focuses on understanding the need for a hydration reminder and designing a program that calculates and reminds users about their daily water intake.

4.1. EXISTING SYSTEM

In the existing system, most people manage their water intake manually. This includes:

- Remembering to drink water without reminders.
- Estimating water consumption without any calculation.
- Using alarms or mobile reminders manually.

➤ Limitations of the Existing System

1. People often forget to drink water regularly.
2. There is no automatic calculation of recommended water intake.
3. Manual reminders may not be consistent.
4. No system to track whether the daily water requirement has been completed.
5. It requires manual effort and discipline from the user.

Because of these limitations, many individuals fail to maintain proper hydration throughout the day.

4.2. PROPOSED SYSTEM

The proposed system is the **Drink Water Reminder System**, which is a simple software application developed in C++.

The system works as follows:

1. The user enters basic details such as **weight and age**.
2. The system calculates the **recommended water intake** based on body weight.
3. The user sets a **reminder interval**.
4. The system automatically reminds the user to drink water after every specified interval.
5. A **beep sound alert** is generated along with the reminder message.
6. The reminders continue until the recommended daily water intake is completed.

➤ Advantages of the Proposed System

- Automatically calculates recommended water intake.
- Provides regular reminders to the user.
- Simple and easy to use.
- Helps users maintain proper hydration.
- Reduces the chances of dehydration.

4.3. BENEFITS OF THE SYSTEM

The Drink Water Reminder System provides several benefits:

- Encourages healthy hydration habits.
- Helps users maintain proper water intake.
- Simple and efficient reminder system.
- Easy to implement and use.
- Can be expanded with additional features in the future.

5.IDENTIFICATION OF NEED

Water is one of the most important elements required for the proper functioning of the human body. It helps in regulating body temperature, transporting nutrients, aiding digestion, and removing waste products from the body. Health experts recommend drinking an adequate amount of water every day to maintain proper hydration. However, in today's modern lifestyle, many people become busy with work, studies, and other activities, which causes them to neglect their daily water intake.

Many individuals do not realize how much water they should consume in a day. The required amount of water varies depending on body weight, age, and physical activity. When people do not drink enough water, it may lead to dehydration. Dehydration can cause several health problems such as fatigue, headaches, dizziness, poor concentration, and reduced physical performance. Therefore, maintaining a proper hydration routine is essential for a healthy lifestyle.

Another major problem is that people often forget to drink water regularly. Even though they know that drinking water is important, they may skip it because they are busy or simply forget. A reminder system can help solve this issue by notifying users at regular intervals to drink water.

The **Drink Water Reminder System** is designed to address these problems by calculating the recommended water intake and reminding users to drink water periodically. This system helps users build a healthy habit of drinking water regularly and ensures that they meet their daily hydration requirements.

❖ Key Needs for the System

- To help people remember to drink water regularly.
- To prevent dehydration and related health problems.
- To calculate the recommended water intake based on body weight.
- To provide automatic reminders at regular intervals.
- To create awareness about the importance of proper hydration.
- To develop a simple and easy-to-use application for everyday use.

6. PRELIMINARY INVESTIGATION

Before developing the system, basic investigation and research were carried out to understand the requirements and feasibility of the project.

The following aspects were studied:

1. Recommended Water Intake Calculation

- ❖ Research shows that a person should drink approximately **0.033 liters of water per kilogram of body weight**.

2. Reminder Mechanism

- ❖ A reminder system can be implemented using **time delays and alerts**.

3. Programming Tools

- ❖ C++ was chosen because it is widely used for learning programming concepts and allows easy implementation of console applications.

4. System Functions

- ❖ Windows provides functions such as `Sleep()` for delays and `Beep()` for alerts, which can be used to create reminder notifications.

The investigation confirmed that the project could be successfully implemented using basic programming techniques.

7.FEASIBILITY STUDY

7.1. TECHNICAL FEASIBILITY

The project is technically feasible because:

- ❖ It uses basic C++ programming concepts.
- ❖ The required libraries and tools are readily available.
- ❖ The program can run on standard Windows computers.

7.2. ECONOMIC FEASIBILITY

The project does not require expensive hardware or software. Free compilers and development tools can be used. Therefore, the project is economically feasible.

7.3. OPERATIONAL FEASIBILITY

The system is easy to use. Users only need to provide basic inputs such as weight and reminder interval. The system automatically performs calculations and provides reminders.

8.PROJRCT PLANNING

Project planning is an important step in the development of any software system. It involves organizing tasks, defining objectives, allocating time, and determining the resources required to complete the project successfully. Proper planning ensures that the project is completed efficiently and within the given time frame.

For the **Drink Water Reminder System**, the project planning process began with identifying the problem and understanding the need for a reminder system that helps users maintain proper hydration. After identifying the problem, the project requirements were analyzed to determine the inputs, processes, and outputs needed for the system.

The next step involved designing the basic structure and logic of the program. During this stage, the system flow was planned, including how the program would accept user input, calculate the recommended water intake, and generate reminders at regular intervals. A suitable formula for calculating water intake was also selected based on body weight.

After the design stage, the coding phase was carried out using the **C++ programming language**. The program was written using basic programming concepts such as variables, loops, conditional statements, and system functions like **Sleep()** and **Beep()**.

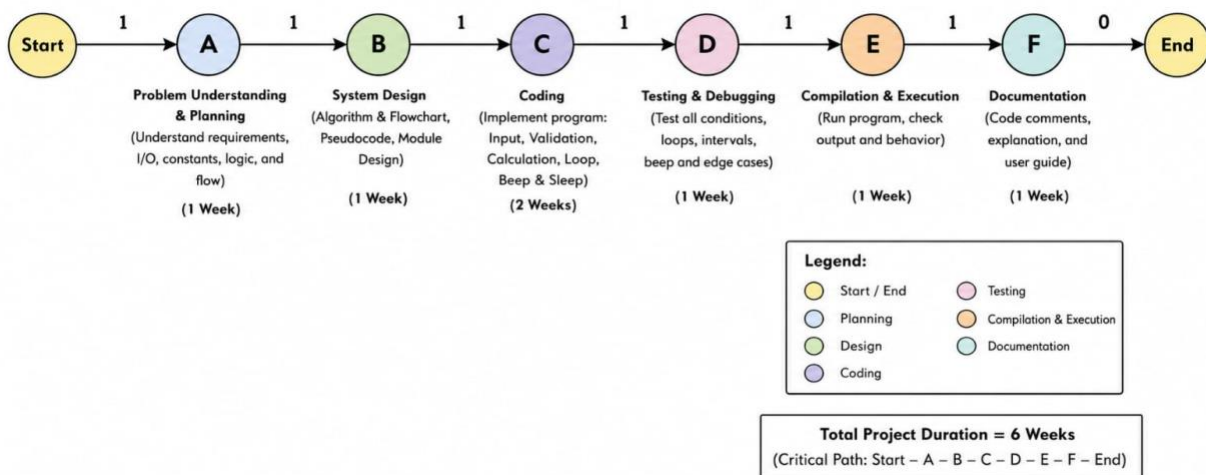
Once the coding was completed, the program was tested to identify and correct any errors. Finally, documentation was prepared to describe the system design, functionality, and overall working of the project.

9. PROJECT SCHEDULING

• GANTT CHART

Task No.	Task / Activity	Duration (Days)	Timeline (Days)														
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	Requirement Analysis	2															
2	Program Design (Flow & Logic)	1															
3	Setting up C++ Environment	1															
4	User Input Module (Weight, Age)	1															
5	Input Validation (Min. Age & Weight)	1															
6	Water Intake Calculation	1															
7	Output Display (Recommended Intake)	1															
8	Reminder Interval Input	1															
9	Reminder Loop with Sleep()	2															
10	Beep Sound Alert	1															
11	Testing & Debugging	2															
12	Final Review & Documentation	1															
Total Estimated Time		15 Days															

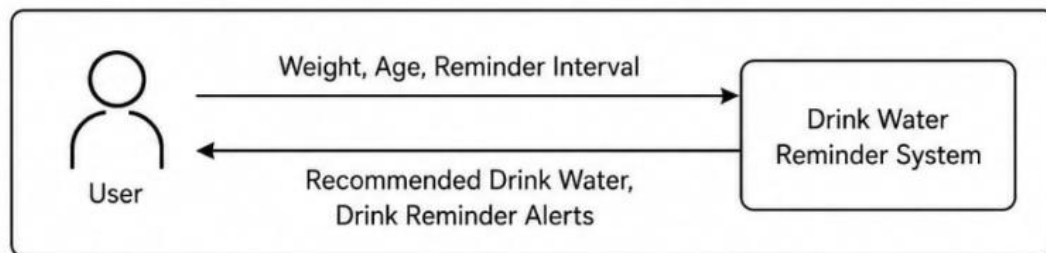
• PERT CHART



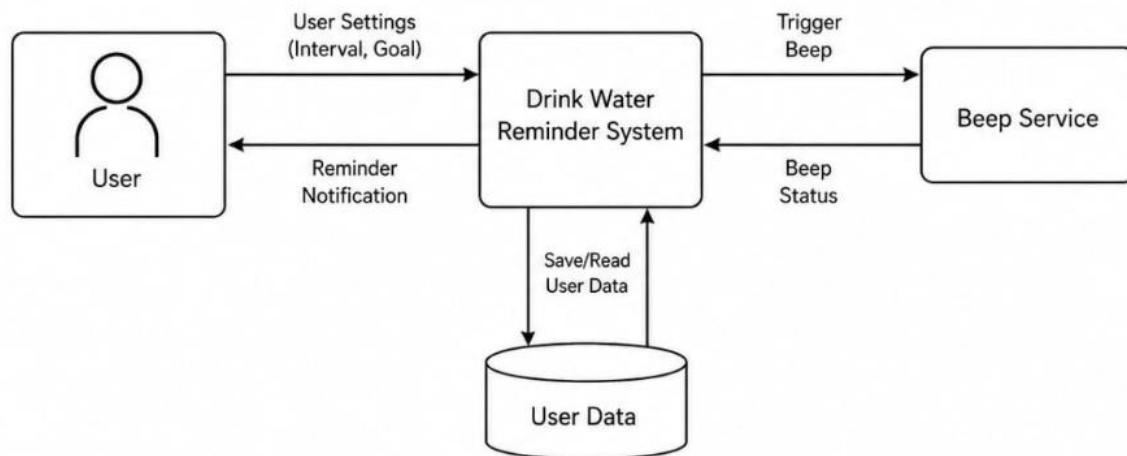
10.DATA FLOW DAIGRAM

A Data Flow Diagram (DFD) is a graphical representation that visualizes how data moves, is processed, stored, and transformed within an information system or business process.

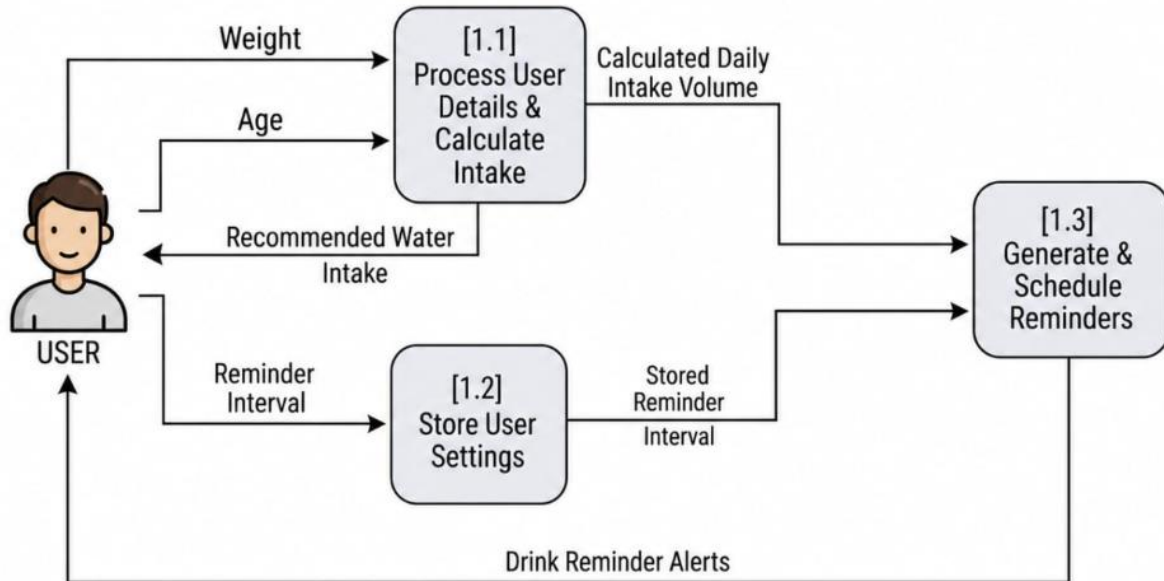
- **0 LEVEL DFD**



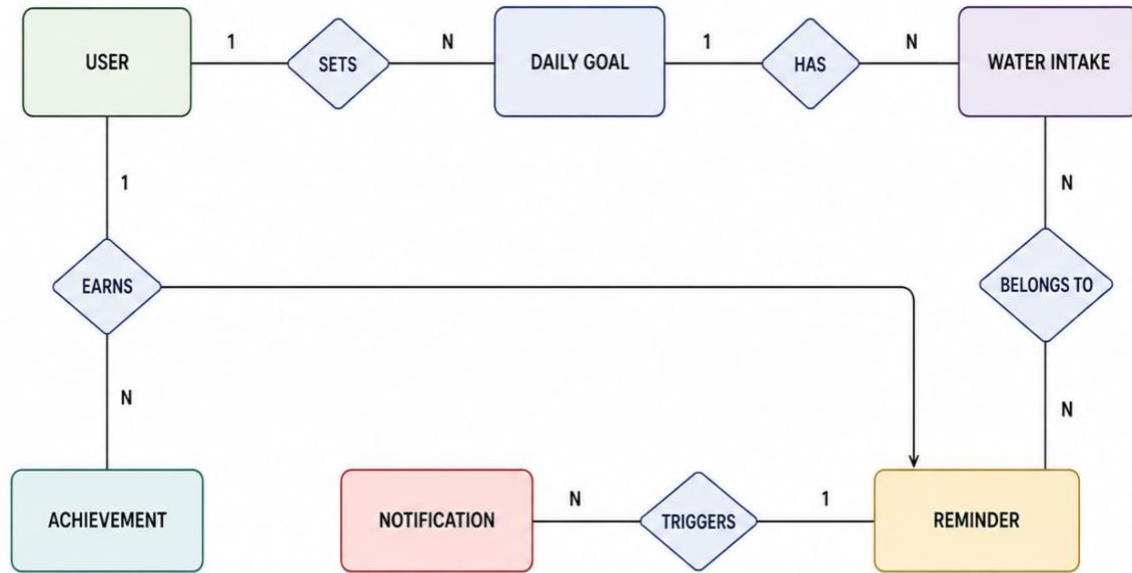
- **LEVEL 1 DFD**



- **LEVEL 2 DFD**



11.ER DAIGRAM



12.CODING

```
#include <iostream>
#include <Windows.h>
using namespace std;

int main()
{
    const double WATER_PER_KG = 0.033;

    double weight, totalWater, remainingWater;
    int age, interval;
    double drinkPerReminder;

    cout << "Enter your weight (kg): ";
    cin >> weight;

    cout << "Enter your age: ";
    cin >> age;

    if (weight < 30 || age < 12)
    {
        cout << "Insufficient data to calculate water intake.\n";
        return 1;
    }

    totalWater = weight * WATER_PER_KG;
    remainingWater = totalWater;

    cout << "\nRecommended daily water intake: " << totalWater << " liters\n";

    cout << "Enter reminder interval (in seconds): ";
    cin >> interval;

    cout << "Enter how much water you will drink each time (liters, e.g. 0.25): ";
    cin >> drinkPerReminder;

    if (interval <= 0 || drinkPerReminder <= 0)
    {
        cout << "Invalid input.\n";
        return 1;
    }

    cout << "\nStarting reminders...\n";

    while (remainingWater > 0.01)
    {
        Sleep(interval * 1000);

        cout << "\nReminder: Drink " << drinkPerReminder << " liters of water.\n";
```

```
    cout << "Remaining: " << remainingWater << " liters\n";

    Beep(750, 500);

    remainingWater -= drinkPerReminder;

    if (remainingWater < 0)
        remainingWater = 0;
}

cout << "\nGreat job! You've completed your daily water intake!\n";

return 0;
}
```

13.OUTPUT

```
C:\Users\MSI\Documents\Unt  ×  +  v
Enter your weight (kg): 75
Enter your age: 19

Recommended daily water intake: 2.475 liters
Enter reminder interval (in seconds): 3
Enter how much water you will drink each time (liters, e.g. 0.25): 0.30

Starting reminders...

Reminder: Drink 0.3 liters of water.
Remaining: 2.475 liters

Reminder: Drink 0.3 liters of water.
Remaining: 2.175 liters

Reminder: Drink 0.3 liters of water.
Remaining: 1.875 liters

Reminder: Drink 0.3 liters of water.
Remaining: 1.575 liters

Reminder: Drink 0.3 liters of water.
Remaining: 1.275 liters

Reminder: Drink 0.3 liters of water.
Remaining: 0.975 liters

Reminder: Drink 0.3 liters of water.
Remaining: 0.675 liters

Reminder: Drink 0.3 liters of water.
Remaining: 0.375 liters

Reminder: Drink 0.3 liters of water.
Remaining: 0.075 liters

Great job! You've completed your daily water intake!

-----
Process exited after 89.25 seconds with return value 0
Press any key to continue . . .
```

14.SYSTEM DESIGN

System design defines how the system is structured and how different components interact with each other to perform the required tasks. In the **Drink Water Reminder System**, the design focuses on creating a simple and efficient program that takes user input, processes the information, and provides reminders to drink water at regular intervals.

The system follows a **simple input–process–output model**. The user first provides input data such as weight, age, and reminder interval. The program then processes this information by validating the input and calculating the recommended water intake based on a predefined formula. After the calculation, the system generates periodic reminders until the recommended water intake is completed.

The design ensures that the system is easy to use and performs its functions accurately. It also ensures that the program runs continuously until the water intake goal is reached.

➤ Input Design

The system requires the following inputs from the user:

- Weight of the user in kilograms.
- Age of the user.
- Reminder interval in seconds.

The program checks whether the weight and age meet the minimum requirements before performing the calculation.

➤ Processing Design

The processing stage includes:

- Validating the user inputs.
- Calculating recommended water intake using the formula:

$$\text{Water Intake} = \text{Weight} \times 0.033$$

- Setting reminder intervals using the **Sleep()** function.
- Displaying reminder messages and generating beep alerts.

➤ **Output Design**

The system produces the following outputs:

- Recommended daily water intake.
- Reminder messages at regular intervals.
- Beep sound alerts.
- Final message after completing the daily water intake.

14.1. Modularization details

Modularization is the process of dividing a program into smaller and manageable parts called modules. Each module performs a specific task, which makes the program easier to understand, maintain, and modify.

In the Drink Water Reminder System, the program can be logically divided into the following modules:

1. Input Module

This module is responsible for collecting information from the user.

Functions performed:

- Accept user weight.
- Accept user age.
- Accept reminder interval.

This module ensures that all necessary data is collected before processing begins.

2. Validation Module

This module checks whether the user input meets the required conditions.

Functions performed:

- Verify that the weight is greater than the minimum weight limit.
- Verify that the age is greater than the minimum age limit.

If the conditions are not satisfied, the program displays an error message and stops execution.

3. Calculation Module

This module calculates the recommended water intake.

Functions performed:

- Apply the formula to calculate water intake.
- Store the calculated value for further processing.

Formula used:

$$\text{Water Intake} = \text{Weight} \times 0.033$$

4. Reminder Module

This module is responsible for generating reminders.

Functions performed:

- Wait for the specified interval using the **Sleep()** function.
- Display reminder messages on the screen.
- Generate beep sound alerts using **Beep()**.

The reminder continues until the water intake goal is reached.

5. Completion Module

This module displays the final message once the recommended water intake is completed.

Functions performed:

- Display a congratulatory message.
- Terminate the program.

14.2. User Interface design

The **User Interface (UI)** is the part of the system that allows users to interact with the program.

In the Drink Water Reminder System, a **console-based user interface** is used.

The interface is designed to be simple and user-friendly so that users can easily enter their information and understand the program output.

➤ Input Interface

The system prompts the user to enter the following details:

- Weight in kilograms
- Age
- Reminder interval in seconds

➤ Output Interface

After receiving the inputs, the system displays the recommended water intake and reminder messages.

15.FUTURE SCOPE

The Drink Water Reminder System can be further improved by adding advanced features. Some possible improvements include:

1. **Graphical User Interface (GUI)** for better user interaction.
2. **Mobile application version** for Android or iOS devices.
3. **Daily water intake tracking system.**
4. **Graphical reports and charts** showing hydration history.
5. **Customizable reminder intervals and water quantity.**
6. **Integration with fitness trackers or smartwatches.**
7. **Notification system using desktop pop-ups or mobile alerts.**
8. **Cloud storage for tracking long-term hydration data.**

These improvements can transform the simple reminder system into a **complete hydration management application.**

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